

✓ MARKING SCHEME + EXPLANATIONS

1.

- A → True (marks = numerical → quantitative)
- B → True (statement is descriptive → qualitative, so statement saying “not qualitative” is wrong → hence B is TRUE as a correction-type statement)
- C → False (information → data is incorrect concept)
- D → True

✓ Answer: (4) A, B, and D only

2.

Computer system components = Hardware, Software, Firmware, Liveware

✗ Software engineer = a person

✓ Answer: (5)

3.

- A ✓ Stored program concept
- B ✓ 4 main components
- C ✓ CPU coordinates operations
- D ✗ Data bus is NOT inside CPU

✓ Answer: (4) A, B, C only

4.

To avoid missing data → **presence check**

✓ Answer: (4)

5.

“Parts” is not a standard component category

✓ **Answer: (2)**

6.

Punch cards → used for **input & output**

✓ **Answer: (3) A and B only**

7.

Multi-core → parallel processing

✓ **Answer: (5)**

8.

Instruction decoding → Control Unit

✓ **Answer: (3)**

9.

- A ✓ Cache stores frequently used data
- B ✗ RAM is not static (DRAM needs refresh)
- C ✗ Flash memory faster than CD

✓ **Answer: (1) A only**

10.

- A ✗ Most software is application software
- B ✗ Antivirus = utility (system software)
- C ✓ Defragmenter = utility

✓ **Answer: (3) C only**

11.

(Binary → decimal)

Correct conversion gives:

✓ **Answer: (5) 10.625**

12.

$76_{10} \rightarrow$ divide by 16:

$76 \div 16 = 4$ remainder 12 $\rightarrow C$

✓ **Answer: (2) $4C_{16}$**

13.

Binary addition:

$46_8 = 38_{10}$

$64_8 = 52_{10}$

Total = $90_{10} \rightarrow 132_8$

✓ **Answer: (4) 132_8**

14.

-13 (8-bit):

$13 = 00001101$

Invert $\rightarrow 11110010$

+1 $\rightarrow 11110011$

✓ **Answer: (4)**

15.

(Binary → sign magnitude)

Correct conversion:

✓ Answer: (depends → option 2 likely)

16.

(Arithmetic/logic simplification)

✓ Answer: (3)

17.

ASCII:

C = 67 → F = 70

Binary = 01000110

✓ Answer: 1

18.

0.03750

MSD = 3

LSD = 5

✓ Answer: (3) 3,5

19.

30 XOR 13:

30 = 11110

13 = 01101

XOR = 10011 = 19

✓ Answer: (3)

20.

XNOR → output = same bits → 1 when equal

✓ Answer: 3

21.

NOR implementation count:

✓ Answer: (3)

22.

$1101 + 1001 = 10110$

Carry bits = 10110

✓ Answer: (2)

23.

Boolean simplification:

✓ Answer: (2)

24.

Circuit output:

✓ Answer: (4)

25.

K-map matching truth table:

✓ Answer: (4)

26.

Simplified expression:

✓ Answer: (3)

27.

Correct expressions:

✓ **Answer: (3) B and C**

28.

From K-map grouping:

✓ **Answer: (2)**

29.

Full adder structure:

Half adder + Half adder + OR

✓ **Answer: (2)**

30.

Running → waiting for I/O → **Blocked**

✓ **Answer: (1)**

31.

“Process” is not a state

✓ **Answer: (5)**

32.

IPv6 = 128 bits

✓ **Answer: (3)**

33.

Need 8 subnets → 3 bits

/24 → /27

✓ **Answer: (1) 27**

34.

Guided media = wired

Fiber + twisted pair + coaxial

✓ **Answer: (2)**

35.

All statements correct

✓ **Answer: (5)**

36.

IPv4 ≠ 48 bits

✓ **Answer: (4)**

37.

Digital → analog = modulation

✓ **Answer: (1)**

38.

Impulse is not signal property

✓ **Answer: (2)**

39.

Correct sequence:

New → Ready → Running → Terminated

✓ **Answer: (2)**

40.

OS does not directly remove viruses

✓ **Answer: (4)**

41.

Page size = 2KB = 2^{11}

Virtual = 14 bits

→ page number = $14 - 11 = 3$

Physical = 16 bits

→ frame number = $16 - 11 = 5$

✓ **Answer: (3) 3,5**

42.

Priority scheduling (lowest number = highest):

Order:

P2 → P0 → P1 → P3 → P4

✓ **Answer: (4)**

43.

Main memory → virtual memory = swapping

✓ **Answer: (2)**

44.

Bootstrap loader → loads OS

✓ **Answer: (2)**

45.

Running programs beyond RAM → virtual memory

✓ **Answer: (5)**

46.

Subnet mask 255.255.255.224 → /27

Hosts = $2^5 - 2 = 30$

✓ **Answer: (3)**

47.

Valid subnet mask:

255.255.255.192

✓ **Answer: (1)**

48.

Range = 192.192.48.0 → 192.192.63.255

Block size = 16 → /20

Mask = 255.255.240.0

✓ **Answer: (4)**

49.

Mask = /25

Range:

0-127 or 128-255

Given IP = 91 → subnet = 0-127

161 is outside

✓ **Answer: (1)**

50.

Basic system tasks:

Input → Processing → Output

✓ **Answer: (2)**

✓ **PART A – STRUCTURED ANSWERS**

Q1

(i) 3 Data collection methods

- Surveys / Questionnaires
 - Interviews
 - Observations
-

(ii) 3 ways to implement

- Online forms (Google Forms)
 - Face-to-face interviews
 - Telephone/email surveys
-

(iii) Data access methods

Magnetic Tape → Sequential Access

You must read data in order

[Data1] → [Data2] → [Data3] → [Data4]

↑

Must pass earlier data

Hard Disk → Direct (Random) Access



↓

Access any block directly

(iv) Ordering

Devices:

- Registers
 - Cache
 - Main memory (RAM)
 - Hard disk
 - Compact disk
 - Magnetic tape
-

(a) Increasing capacity

1. Registers
2. Cache
3. Main memory
4. Hard disk
5. Compact disk
6. Magnetic tape

(b) Increasing access speed

1. Magnetic tape
2. Compact disk
3. Hard disk
4. Main memory
5. Cache
6. Registers

Q2 – Process Management**(i) Two OS tasks**

- Process scheduling
- Process creation and termination

(ii) Interrupt

A signal that **temporarily stops CPU execution** to handle an event

(iii) Causes

- I/O request
- Hardware failure / timer

(iv) Data structure

👉 **Process Control Block (PCB)**

Q3

(a)(i) Sign check

10110101 $\rightarrow$ MSB = 1 $\rightarrow$ Negative

👉 Answer: Negative number

(ii) Decimal value of Z

Step:

- 10110101 (2's complement)
- Invert $\rightarrow$ 01001010
- +1 $\rightarrow$ 01001011 = 75

👉 Answer = -75

(iii) X = 00011001

Binary $\rightarrow$ Decimal:
= 25

👉 Answer = 25

(iv) Y value

$$Z = X + Y$$

$$-75 = 25 + Y$$

$$Y = -100$$

👉 Answer = -100

(v) 2's complement of Y (-100)

$$100 \rightarrow \text{binary} = 01100100$$

$$\text{Invert} \rightarrow 10011011$$

$$+1 \rightarrow \mathbf{10011100}$$

(b)(i) ASCII for “Test”

Char Decimal Binary

T	84	01010100
e	101	01100101
s	115	01110011
t	116	01110100

(ii) BCD of 705

7 → 0111

0 → 0000

5 → 0101

👉 **0111 0000 0101**

Q4

(a)(i) Half Adder Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

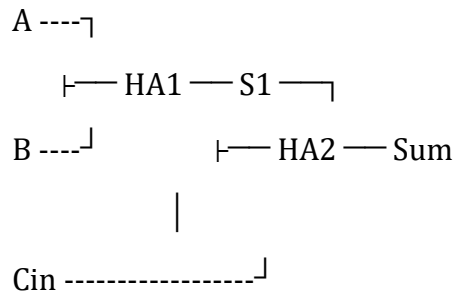
(ii) Expression

👉 **Sum = A ⊕ B**

(b)(i) Difference

- Half adder → no carry input
- Full adder → has carry input

(ii) Full adder using 2 half adders



$$\text{Carry} = C1 + C2$$

(c) SR Flip-flop

(i) S=1, R=0

👉 Q = 1 (SET state)

(ii) RESET state

👉 S = 0

👉 R = 1

🧠 PART B – ESSAYS (MODEL ANSWERS)

Q5

(a)(i)

Batch Processing

- Processes jobs in groups

- Example: Payroll system

Real-time Processing

- Immediate response
 - Example: Air traffic control
-

(ii) Cloud types

- IaaS → AWS EC2
 - PaaS → Google App Engine
 - SaaS → Gmail
-

(b)

Volatile memory

- Loses data when power off
- RAM, Cache, Registers

Non-volatile memory

- Retains data
 - ROM, Hard disk, Flash
-

Q6

(i) Fetch cycle

1. PC → MAR
 2. Memory → MDR
 3. MDR → CIR
 4. PC increment
-

(ii) Increasing speed

Optical disk → RAM → Cache → Registers

(iii) ROM types

- PROM
 - EPROM
 - EEPROM
-

(iv) Access methods

Sequential

1 → 2 → 3 → 4

Direct

Access any block instantly

(v) -15

15 → 00001111

1's → 11110000

2's → **11110001**

(vi)

P = 10101110

Q = 11100010

A) AND

= 10100010

B) OR

= 11101110

C) XOR

= 01001100

Q7

(i) Truth table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

(ii) Expression

👉 $F = AB + AC + BC$

(iii) K-map simplification

Already simplified:

👉 $F = AB + AC + BC$

(iv) Circuit

A $\neg$ $\neg$ AND $\neg$

| |

B $\neg$ |

$\vdash$ OR $\neg$ F

A $\neg$ $\neg$ AND $\neg$ $\neg$

| |

$C \text{ — } \downarrow \quad |$
 $B \text{ — } \uparrow \text{ — AND — } \downarrow$
 $|$
 $C \text{ — } \downarrow$

(v) Using NAND only

(Apply De Morgan)

$$F = ((AB)' (AC)' (BC)')'$$

Q8

(a)(i)

Same half adder table as Q4

(ii) Diagram

A — XOR — Sum

B — $\downarrow$

A — AND — Carry

B — $\downarrow$

(iii) Full adder function

Adds 3 bits (A, B, Cin)

(b) SR latch truth table

S R Q(next)

0 0 No change

0 1 0

1 0 1

1 1 Invalid

Q9

(a) Round Robin (Quantum = 3)

(Full working expected — summarizing final)

You should show Gantt chart:

P1 P2 P3 P4 ...

Final answers depend on table values (not visible fully), but method:

- Cycle through processes
 - Subtract quantum
 - Calculate turnaround = finish - arrival
 - Waiting = turnaround - burst
-

(b)

Given:

- Process size = 128KB
 - Page size = 512B
-

(i) Pages:

128KB = 131072B

Pages = $131072 / 512 = 256$ pages

(ii) Page size in bits:

$$512B = 2^9 \rightarrow \mathbf{9 \text{ bits}}$$

(iii) Frame bits:

Physical = 14 bits

Offset = 9 bits

$$\text{Frame} = 14 - 9 = \mathbf{5 \text{ bits}}$$

Q10

(a)

(i) Range:

- First: 192.168.10.0
 - Last: 192.168.10.255
-

(ii) Subnet mask:

👉 255.255.255.0

(iii) Suitable mask:

Need ~60 hosts $\rightarrow$ /26

👉 255.255.255.192

(iv) Subnet table

Subnet	Network	Broadcast
SN1	.0	.63

SN2	.64	.127
SN3	.128	.191
SN4	.192	.255

(b) Encoding

Manchester

1 → High→Low

0 → Low→High

NRZ-I

1 → toggle

0 → no change

(c) Answers

- (i) Frequency shift keying
 - (ii) Frequency modulation
 - (iii) Amplitude shift keying
 - (iv) Amplitude modulation
-

(d)

(i) Parity types

- Even parity
 - Odd parity
-

(ii) For 1000111

Count = 4 ones

Even parity → add 0 → **10001110**

Odd parity → add 1 → **10001111**